

Rate-Risk Relationship

L. Childs

May 30, 2026

Consider a population x modeled in continuous time with mortality rate a such that the decline of the population is model by

$$\frac{dx}{dt} = -ax,$$

whose solution is

$$x(t) = x(0)e^{-at}.$$

At time $t + 1$, the solution is

$$x(t + 1) = x(0)e^{-a(t+1)}, \tag{1}$$

and at time t , the solution is

$$x(t) = x(0)e^{-at}. \tag{2}$$

Consider the same population x modeled in discrete time with mortality risk α such that the decline of the population is model by

$$x(t + 1) = x(t) - \alpha x(t). \tag{3}$$

Plugging Eq. 1 and Eq. 2 into Eq. 3, we find

$$x(0)e^{-a(t+1)} = (1 - \alpha)x(0)e^{-at}.$$

Canceling out $x(0)$ and e^{-at} from both sides, we are left with

$$e^{-a} = 1 - \alpha,$$

and rearranging gives

$$\alpha = 1 - e^{-a}.$$

In other words,

$$\text{risk} = 1 - e^{-\text{rate}}.$$